

When the Loop Closes Quietly

Mechanistic Interpretability as Contestability Infrastructure for Deployed AI

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Paper + references

Every metric said the deployment worked. Overrides had fallen to near zero.
Contestability had degraded — and nothing in the monitoring framework could see it.

1 The closed loop

Six counties, 2024–2025. Field observation, not instrumented measurement.

An AI decision-support system fused geophysical, satellite and geospatial data into pipelines for subsurface anomaly detection. Users were county officials deciding land use and permitting; the affected public was artisanal miners.

What the metrics caught

Weeks of processing became hours. Excavation failures and wasted survey effort fell. An honest monitoring framework would have called it a success.

What they missed

Officials stopped contesting outputs they could not evaluate. Citizens could no longer trace a permitting decision to a human reasoning process. The official cited the system. The system cited its training. The training was elsewhere.

Early on, officials overrode, queried and annotated flags. By the final quarter overrides were near zero. Not measured systematically, not offered as a metric — but unambiguous to everyone running the deployment.

The claim is narrow. Not that the system was wrong or officials negligent. That the conditions under which an output *could* be contested had degraded, invisibly to every metric built to check whether it was working.

2 One traced decision

What contestability infrastructure actually buys.

LOOP CLOSED

A parcel is flagged at confidence 0.83. A community contests: the parcel is under customary claim. The official cites the system. The flag stands.

LOOP REOPENED

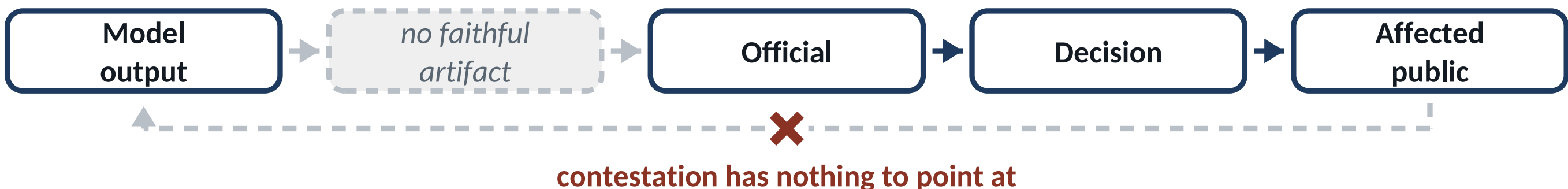
A feature attribution for that output shows the flag fired on surface reflectance correlated with prior excavation, not the subsurface conductivity the system was built to detect. The model learned that disturbed ground predicts viable sites: extraction history, not geology.

The contestation now has an artifact to point at, the artifact has a defect anyone trained can name, and the outcome is no longer “the model said so.”

Why not SHAP or LIME?

A local surrogate approximates the model, so it yields a plausible account of the computation, not the computation. For most uses that gap is academic. For contestation it is the whole problem: an unfaithful explanation supplies a traceable-looking reason while concealing the real one.

LOOP CLOSED



LOOP REOPENED

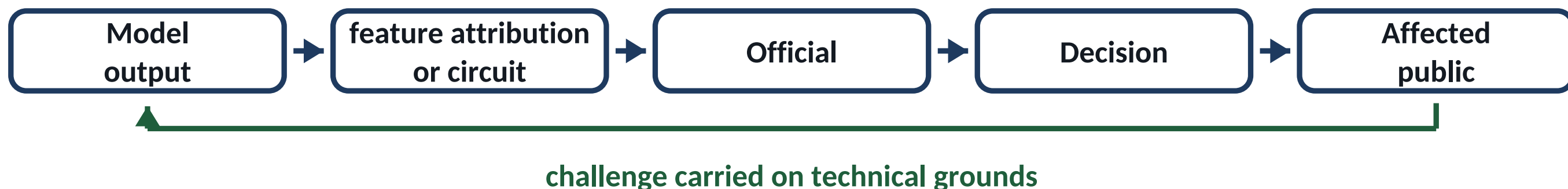


Figure 1. The same decision chain with and without a faithful artifact. The break is in the return path, not the forward one — the system works as designed either way.

3 Failure modes map to primitives the field already has

Illustrative, not exhaustive. The claim is not that mech interp solves contestability, but that contestability has technical preconditions and this is the field building them.

Contestability gap	Failure mode	Interpretability primitive
Untraceable decisions	Output pathway cannot be explained	Circuit analysis
Uncontested outputs	Model reasoning cannot be evaluated	Feature attribution
Confounded signals	Output rests on the wrong evidence source	Sparse autoencoders
Hidden degradation	Distribution shift goes unnoticed	Activation monitoring

4 What it cannot solve

Standing. An explanation helps only someone entitled to challenge. The communities most affected had no procedural access at all.

Capacity. Artifacts need readers: a civil society technical team, an ombudsperson. Thinnest where loops close fastest.

Access. These methods presume weights or activations. A government running a vendor model has neither.

Values. Why an output happened does not settle whether it was right.

Bad faith. Interpretable systems still close loops when artifacts are withheld.

Without it, standing, capacity, access and procedure operate over an empty technical core.

5 What changes

Research priorities

Science values novelty, generality, depth. Infrastructure values reliability under deployment, accessibility to non-specialists, queryability by low-capacity institutions, stability across updates. The second set is undersupplied.

Governance asks

If contestability requires internals, access to internals is a regulatory question: activation logging, weight access for designated auditors, feature documentation for high-stakes deployments.

Relevant publics

Civil society, affected communities, public-sector deployers, courts, ombudspersons, journalists. None appear in the field's communicative imagination. They should.

6 Four predictions

- Override rates track interpretability access, not performance: steeper decline where access is lower, independent of accuracy.
- Deference scales with opacity. Hold accuracy fixed, vary explanation availability.
- Contestation decouples from error: challenges fall even as performance improves.
- Faithful artifacts raise successful challenge rates against matched deployments without access.

If override rates track performance, or deference is flat across opacity, the proposed mechanism is not doing the work.

None needs frontier-scale infrastructure. One comparative deployment study, or one operator experiment, reaches all four.